

THE UNIVERSITY OF AZAD JAMMU AND KASHMIR, MUZAFFARABAD

Department of Software Engineering

Lab 02:

Course: Software Construction & Development

(Lab)

Semester: Fall 2026

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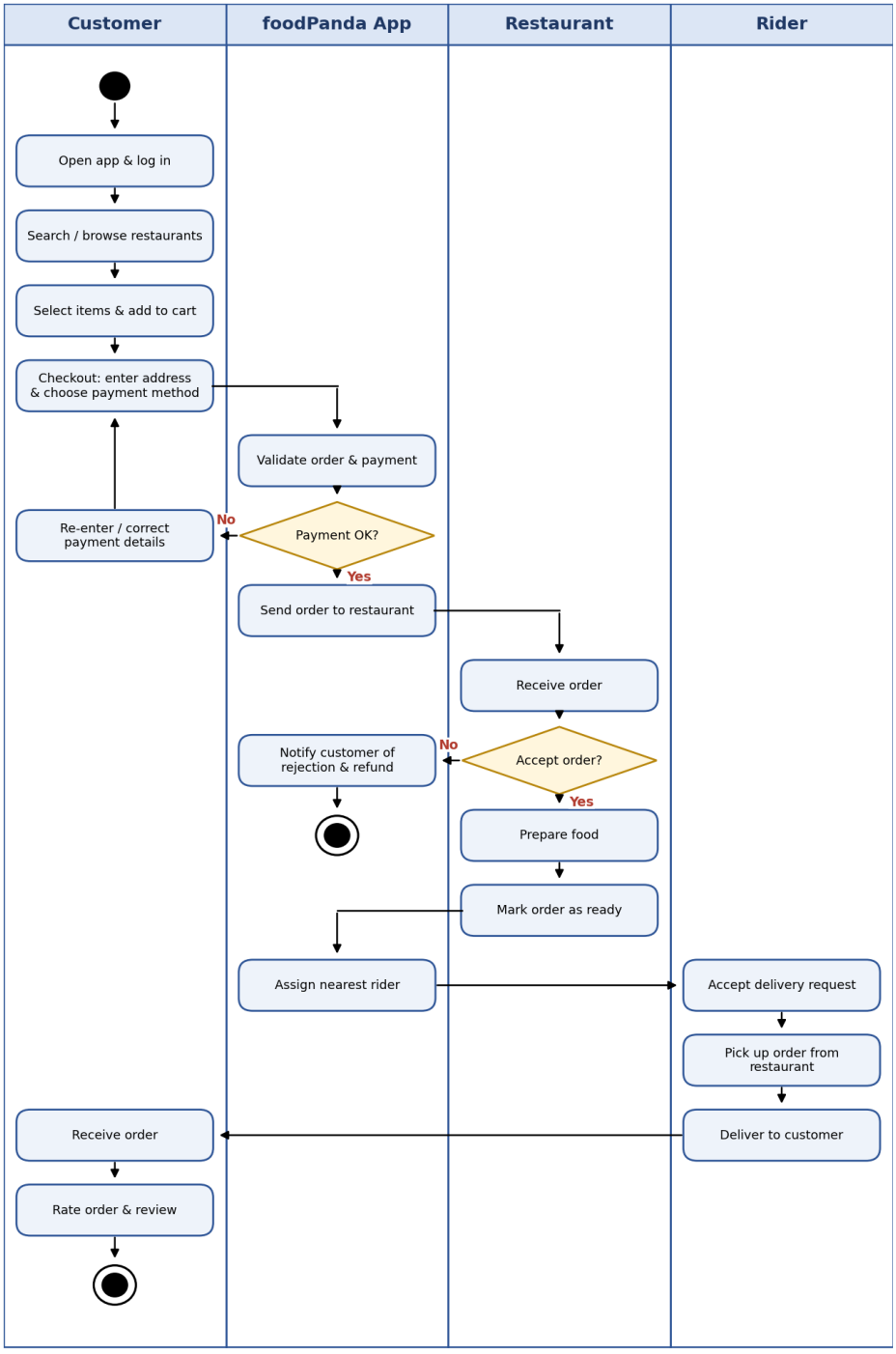
Course Code: SE-315L / CS-315L

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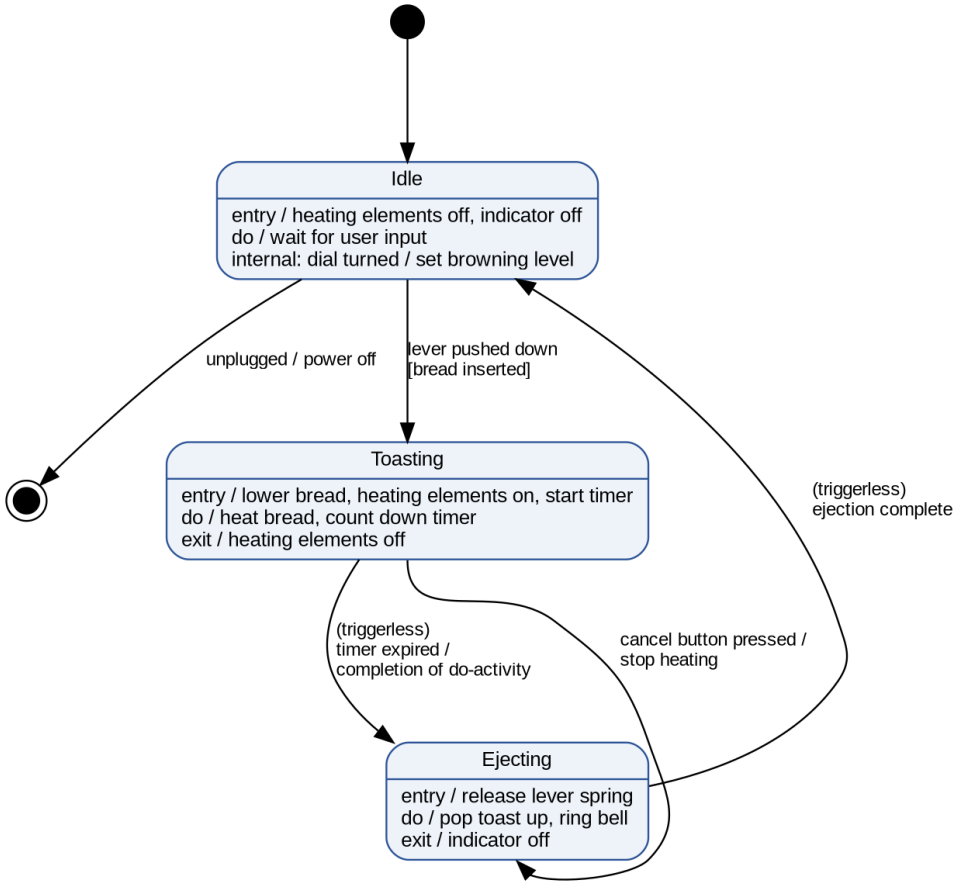
Lab 2: Activity and State Machine Diagrams

Q1: foodPanda Activity Diagram with Swimlanes



Model solution for Lab 2, Q1

Q2: Toaster State Diagram



Model solution for Lab 2, Q2

States and activities

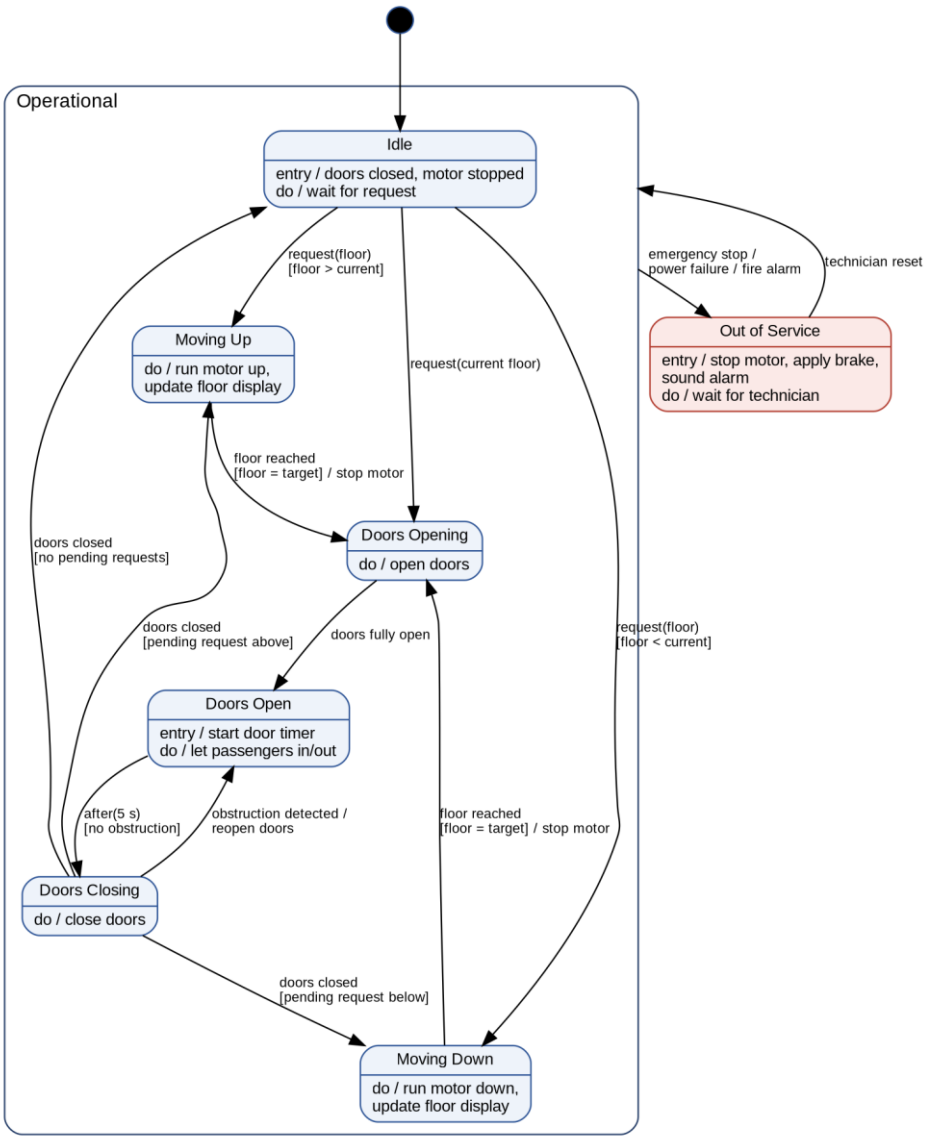
State	Entry	Do	Exit
Idle	Heating elements off, indicator off	Wait for user input	-
Toasting	Lower bread, elements on, start timer	Heat bread, count down timer	Heating elements off
Ejecting	Release lever spring	Pop toast up, ring bell	Indicator off

Transitions

Transition	Event / action	Type
Idle → Toasting	Lever pushed down [bread inserted]	Triggered
Toasting → Ejecting	Timer expires (do-activity completes)	Triggerless
Toasting → Ejecting	Cancel button pressed / stop heating	Triggered
Ejecting → Idle	Ejection complete	Triggerless
Idle → final	Unplugged / power off	Triggered
Idle (internal)	Dial turned / set browning level	Triggered, no state change

A triggered transition fires on an explicit event; a triggerless (completion) transition fires automatically when the state's do-activity finishes.

Q3: Elevator / Lift State Machine Diagram



Model solution for Lab 2, Q3